

CDFD Runtime Paper 06: CLI-First Platform and Future Web Architecture

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May 2026

Abstract

Executive synthesis. The platform direction is CLI first, web app second. The CLI is for serious engine work: reproducible simulations, paper diagnostics, debugging, automation, and saved results. An optional web studio (**webapp/**) is a visual layer over the same backend, not a separate physics system. This paper defines that boundary and explains how CLI outputs flow into optional visual interfaces.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α , β , and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or re-distribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η , λ , μ , ν , ρ , σ , τ , and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers

therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Ordering Principle

The platform order is:

1. core engine;
2. CLI;
3. API server;
4. web app.

The current repository has the engine and CLI as primary surfaces. The initial web studio under `webapp/` (CDFD Runtime Studio, Streamlit) is optional: it is a visual layer over the same kernel, `runtime/diagnostics.py`, and domain adapters and does not duplicate physics logic.

3 CLI Responsibilities

The CLI supports:

- listing registered domains (`cdfd.py domains`);
- Part II diagnostics export (`cdfd.py diagnostics`);
- running domain demos with JSON payloads or OOL source scenarios;
- validating and executing CDFL models;
- exporting JSON result envelopes;
- app-key boundary checks (`cdfd.py auth`);
- preserving finite-value and provenance metadata on every command.

This is the credibility layer. It gives papers exact commands instead of vague claims that "the engine was run."

4 CDFD Runtime Studio

The public Streamlit studio (`python -m webapp.run_server`) already provides an initial visual environment: regime-banded trajectories, field heatmaps, Part II source-mix panels, a Life Number calculator, Granger-style causal graphs from run history, and the `origins_of_life` domain explorer. Future CDFD Studio releases may add model building, coherence monitoring, and deeper result browsing on top of the same CLI backend.

The first public demos remain non-medical and visual:

- heat diffusion and thermal flow;

- traffic or infrastructure routing;
- resource-flow networks;
- predator-prey/ecology models;
- photonic or adaptive-surface demos.

Medical applications require stricter validation, privacy, and user-safety boundaries and are not the first public web surface.

5 API Boundary

The API exposes command-like operations:

- validate a CDFL model;
- run a CDFL model;
- retrieve saved result envelopes;
- inspect finite audits;
- list experiments and selected outputs;
- export reports.

Each endpoint calls `runtime/runner.py` or a later service wrapper around it.

6 API Keys and the AI Boundary

Application keys belong at the application boundary, not inside the physics kernel. The runtime can check whether a calling app presents an approved key, record that key as a redacted fingerprint in provenance, and then run CDFL or domain commands. It does not call an LLM, store LLM credentials, or generate application-facing responses. That layer belongs in CDFD Studio, regulated apps, research notebooks, or other governed products built above the engine.

This split keeps the engine clean: `cdfd.py` and `runtime/runner.py` own deterministic execution, finite audits, and result envelopes; applications own authentication, user identity, LLM routing, rate limits, privacy, and workflow policy.

7 Vacuum OS

Vacuum OS (VOS) is the proposed operating system layer above CDFD Runtime. VOS is not a second engine. It is the orchestration shell for projects that need apps, CDFL packages, user spaces, experiment queues, provenance stores, and controlled AI assistance. In this architecture, CDFD Runtime remains the mathematical kernel; VOS becomes the operating environment that launches models, manages keys, schedules runs, stores outputs, and connects approved applications to the same CDFD command backend.

8 Experiment Visualization

The experiment outputs already define future panels:

- vacuum hysteresis: M_s residual heatmap and recovery curve;
- knot stability: stability-index trace toward the χ^* attractor;
- OOL phase: $\Lambda = 1$ phase boundary;
- frontier sweep: pattern distribution and candidate triage table;
- gigamarathon sweep: domain coverage dashboard.

The public Streamlit dashboard under `webapp/dashboard.py` already implements an initial subset: regime-banded Ψ_s trajectories, Φ and C field heatmaps, Part II aromatic source-mix bars, a Life Number calculator, Granger-style causal edges from run history, and the `origins_of_life` domain explorer. The web app presents these as model diagnostics and falsification targets, not as empirical proof.

9 Conclusion

The platform makes one thing unmistakable: CDFD is executable. The CLI establishes that discipline. The optional web studio makes the same discipline visible without weakening the runtime boundary or replacing command-line reproducibility.

References

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